



University of Piraeus
School of Information and Communication Technologies
Department of Digital Systems

Level: Master's Programme of Studies

Cyber Defence and Digital Forensics

Assignment 1

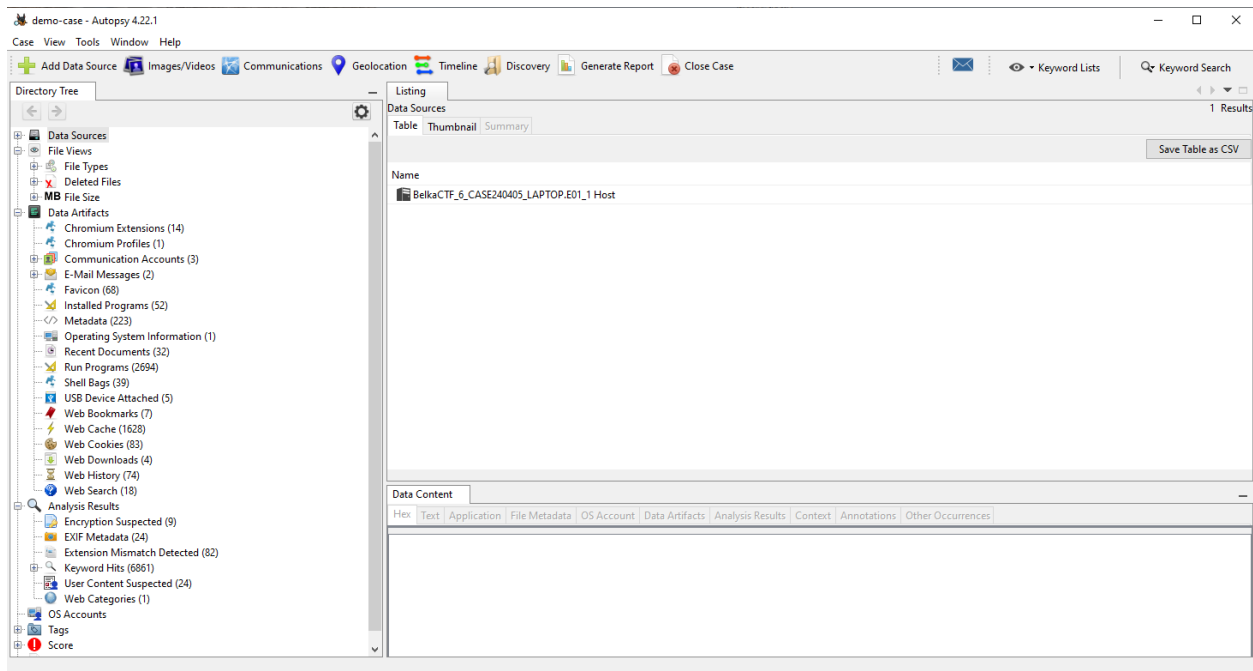
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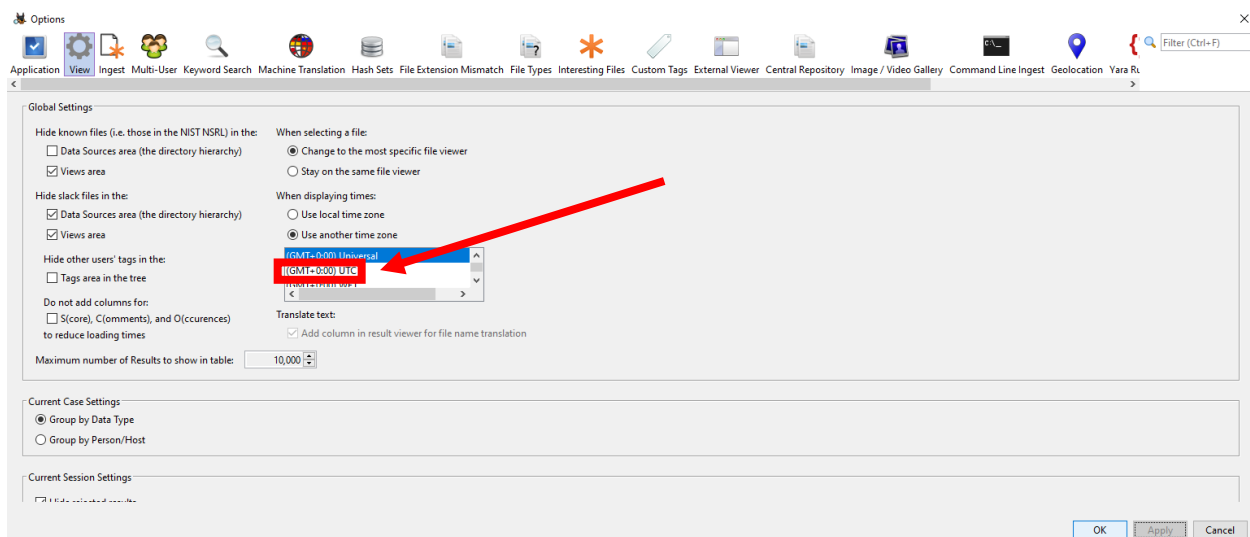
Piraeus

The instructions for installing Autopsy and preparing for answering all questions were followed.



Q1: When was the current version of the operating system installed on this system? (If you used Autopsy to find the answer add a ‘Bookmark’ tag to the artifact that supports your answer- Provide the timestamps in UTC):

Note: If the timezone in Autopsy is not set to UTC, you can change timezone by clicking on Tools->Options->View and change the timezone to UTC. Then click on ‘OK’ to apply.



Answer:

After first setting the correct time zone in Autopsy to UTC (Tools -> Options -> View), we proceeded to find the OS installation date.

Initially, the search was carried out in the automated 'Operating System Information' section, without finding the information there.

Opening the 'SOFTWARE' file in the 'Application' tab, we navigated to Windows -> Windows NT -> CurrentVersion. There we found the value of the 'InstallDate' key.

This value is in Unix Timestamp (or Epoch time) format, representing the number of seconds elapsed since 01/01/1970.

The OS installation date is 31 March 2022, at 21:49:57 UTC.

Q2: Answer the following questions regarding the user with SID “S-1-5-21-1548112901-4150280657-3748420248-1000” (If you used Autopsy to find the answer add a ‘*Bookmark*’ tag to the artifact that supports your answer- Provide the timestamps in UTC):

- a) What is the username of this account?
- b) When was the last login performed?
- c) When was the password last changed?

Answer:

- a) We find the category named 'OS Accounts' and left-click on it. In the top-right window (Listing), the account phorger appears.

Listing 11 Results

Table Thumbnail Summary

Save Table as CSV

Name	S	C	O	Login Name	Host	Scope
S-1-5-80-956008885-3418522649-1831038044-185329			0		BelkaCTF...	Local
S-1-5-18				SYSTEM	BelkaCTF...	Local
S-1-5-21-1548112901-4150280657-3748420248-1000			0	phorger	BelkaCTF...	Domain
S-1-5-80-3028837079-3186095147-955107200-370196			0		BelkaCTF...	Local
S-1-5-19				LOCAL SERVICE	BelkaCTF...	Local
S-1-5-20				NETWORK SERVICE	BelkaCTF...	Local
S-1-5-21-3933942852-973373972-2766786355-1032			0		BelkaCTF...	Domain
S-1-5-21-1548112901-4150280657-3748420248-504			0	wdagutilityaccount	BelkaCTF...	Domain

< >

Data Artifacts	Analysis Results	Context	Annotations	Other Occurrences
Hex	Text	Application	File Metadata	OS Account

Basic Properties

Login: phorger

Full Name:

Address: S-1-5-21-1548112901-4150280657-3748420248-1000

Type:

Creation Date: 2022-03-31 21:49:50 UTC

Object ID: 429

BelkaCTF_6_CASE240405_LAPTOP.E01_1 Host Details

Last Login: 2024-12-27 02:06:49 EET

Login Count: 15

Password Settings: Password not required

Flag: Normal user account

Home Directory: C:/Users/phorger

Realm Properties

Name: Unknown

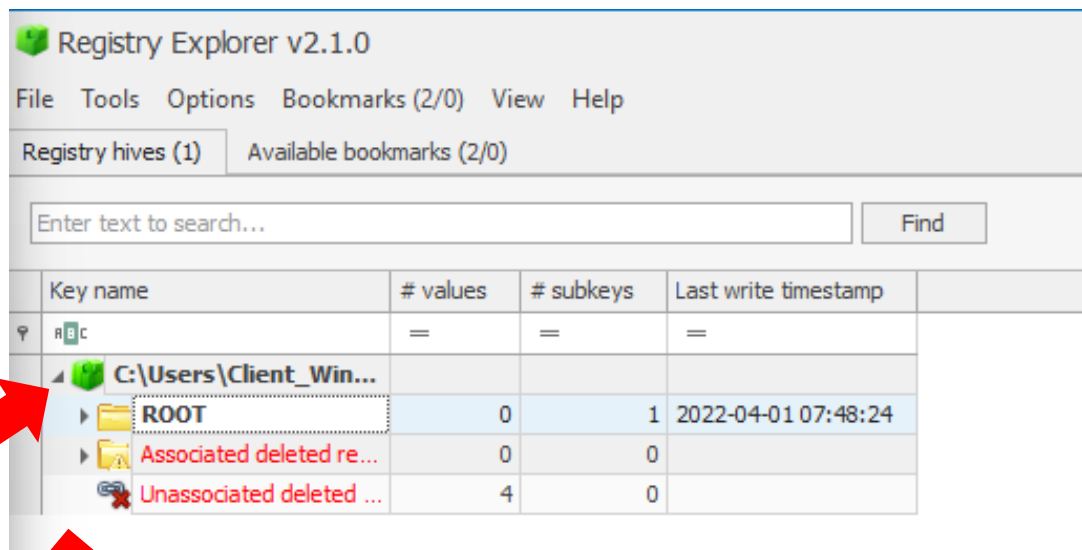
Address: S-1-5-21-1548112901-4150280657-3748420248

Scope: Domain

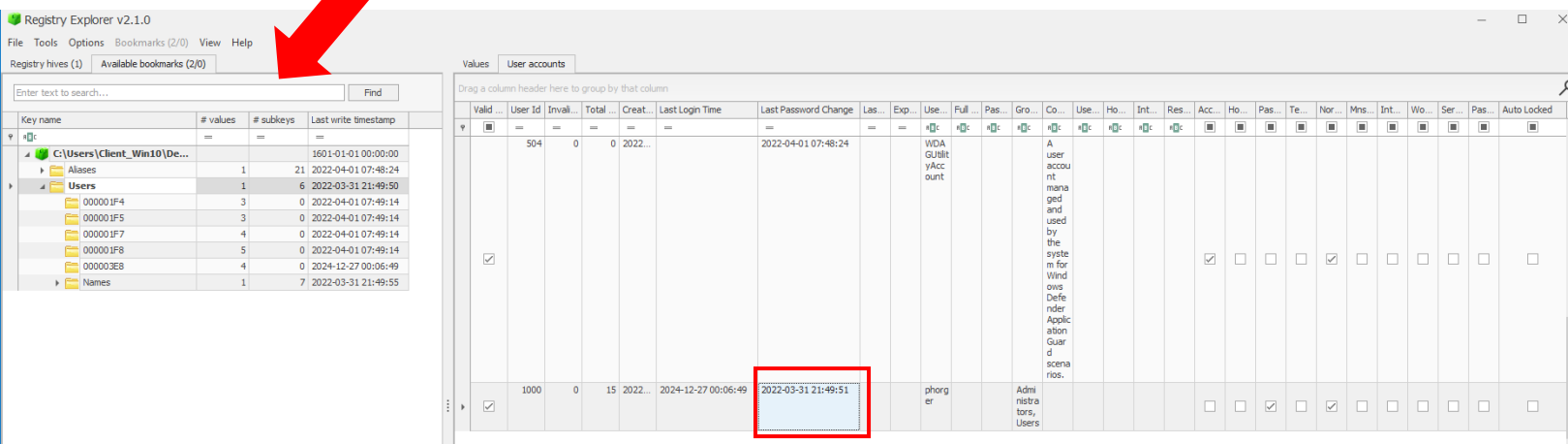
Confidence: Known

To username είναι το **phorger**.

- b) In the 'Host Details' section we see the value Last Login: 2024-12-27 02:06:49 EET. For some reason it displayed this time zone instead of UTC.
- c) In this specific image, there is no information about the date of the last password change. Therefore, we navigate to the Registry Explorer tool.



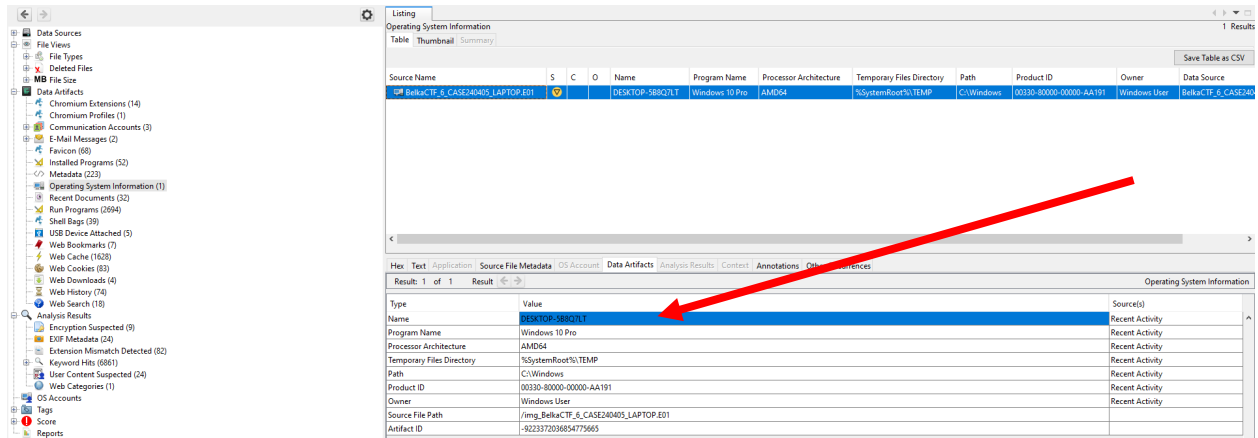
Για τον χρήστη phorger (με User Id 1000), η στήλη «Last Password Change» γράφει: 2022-



03-31 21:49:51.

Q3: What is the computer name of this system? (If you used Autopsy to find the answer add a 'Bookmark' tag to the artifact that supports your answer)?

Answer:



Within the Autopsy environment, in the left menu (Directory Tree), the sub-category 'Operating System Information' was selected.

Q4: Answer the following questions regarding the user “*phorger*”, who searched the web for specific instructions related to anti-forensics (If you used Autopsy to find the answer add a ‘*Bookmark*’ tag to the artifact that supports your answer- Provide the timestamps in UTC):

- What was the search query?
- When did the user perform this search?
- Which web browser did the user use?
- Which search engine did the user use?

Answer:

- Via Autopsy, in the left menu the sub-category 'Web Search' was selected, where the search history is recorded.

From the above image it emerges that:

- The search date is 2024-01-03 01:19:15 UTC.
- The user used the Microsoft Edge browser.
- The search engine is bing.com.

Q5: Answer the following questions regarding the downloaded items of the user “phorger” (If you used Autopsy to find the answer add a ‘*Bookmark*’ tag to the artifact that supports your answer- Provide the timestamps in UTC):

- a) The user appears to have downloaded a ZIP archive from the Internet. Within this archive, there is one executable file. Can you calculate its SHA-256 hash value?
- b) The same executable can be found in the user’s Desktop folder. Do these two executables have the same SHA-256 hash value?

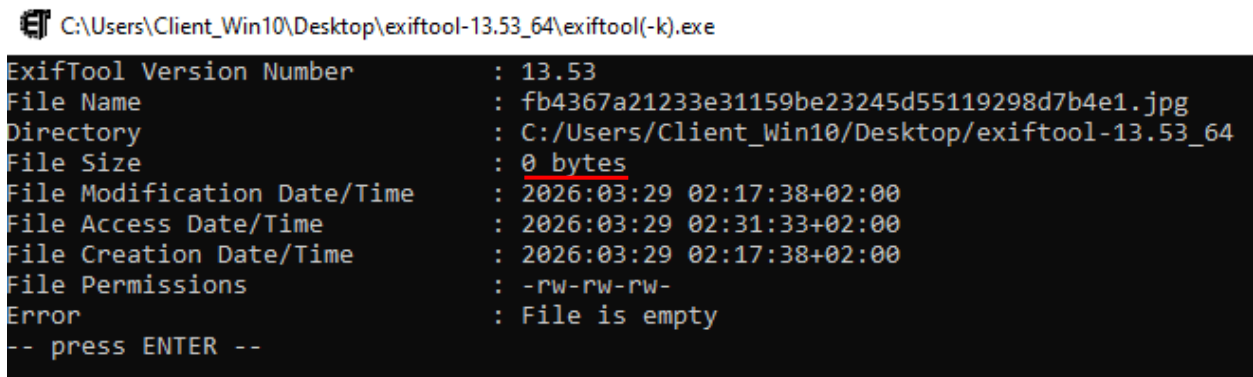
Answer:

- a) In the first row of the table we can clearly see the file powder.zip, which was downloaded to the folder C:\Users\phorger\Downloads\.
- b) Via the Directory Tree at the path vol6\Users\phorger\Desktop, the corresponding executable file named Powder.exe was located.

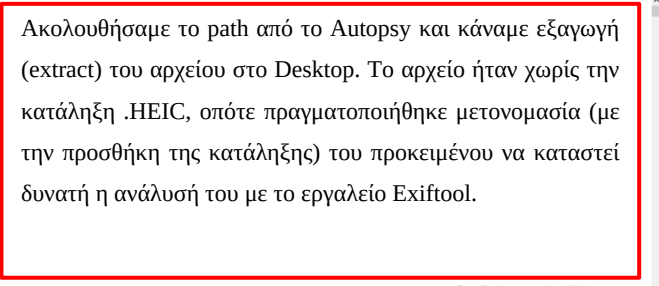
Q6: Locate the file with the MD5 hash value “0ee7490fedb9247af9e5ff64042f3714”. Can you determine the amount in euros (EUR) that the user appears to have paid for Airbnb services?

Answer:

Using Autopsy's 'File Search by Attributes' tool and searching based on the above MD5 hash, the file was located.



This difficulty was due to the fact that Autopsy was trying to retrieve a JPG preview from a previous folder.



C:\Users\Client_Win10\Desktop\exiftool-13.53_64\exiftool(-k).exe

```
ExifTool Version Number      : 13.53
File Name                    : fb4367a21233e31159be23245d55119298d7b4e1.heic
Directory                   : C:/Users/Client_Win10/Desktop/exiftool-13.53_64
File Size                    : 1953 kB
File Modification Date/Time  : 2024:03:18 20:53:22+02:00
File Access Date/Time       : 2026:03:29 04:07:36+03:00
File Creation Date/Time     : 2024:03:18 20:53:12+02:00
File Permissions             : -rw-rw-rw-
File Type                    : HEIC
File Type Extension         : heic
MIME Type                   : image/heic
Major Brand                  : High Efficiency Image Format HEVC still image (.HEIC)
Minor Version                : 0.0.0
Compatible Brands            : mif1, MiPr, miaf, MiHB, heic
Handler Type                 : Picture
Primary Item Reference       : 49
Meta Image Size              : 4032x3024
Exif Byte Order              : Big-endian (Motorola, MM)
Make                         : Apple
Camera Model Name            : iPhone 8
Orientation                  : Rotate 90 CW
X Resolution                  : 72
Y Resolution                  : 72
Resolution Unit               : inches
Software                     : 16.3
Modify Date                   : 2023:11:09 09:38:58
Host Computer                 : iPhone 8
Exposure Time                 : 1/50
F Number                      : 1.8
Exposure Program              : Program AE
ISO                           : 160
Exif Version                  : 0232
Date/Time Original            : 2023:11:09 09:38:58
Create Date                   : 2023:11:09 09:38:58
Offset Time                   : -06:00
Offset Time Original          : -06:00
Offset Time Digitized         : -06:00
Shutter Speed Value           : 1/50
Aperture Value                : 1.8
Brightness Value              : 2.417279939
Exposure Compensation         : 0
Metering Mode                 : Multi-segment
Flash                         : Off, Did not fire
Focal Length                  : 4.0 mm
Subject Area                  : 2015 1511 2217 1330
Maker Note Version            : 14
Run Time Flags                : Valid
Run Time Value                : 2805537796500
Run Time Scale                : 1000000000
Run Time Epoch                : 0
AE Stable                     : No
AE Target                     : 195
AE Average                    : 205
```

```

Focal Length In 35mm Format : 28 mm
Lens Info : 3.99mm f/1.8
Lens Make : Apple
Lens Model : iPhone 8 back camera 3.99mm f/1.8
Profile CMM Type : Apple Computer Inc.
Profile Version : 4.0.0
Profile Class : Display Device Profile

```

- The camera model is iPhone 8.
- The device orientation at the time of capture was Rotate 90 CW (clockwise rotation by 90 degrees).
- The lens model is iPhone 8 back camera 3.99mm f/1.8.
- The photograph was originally created at 2023:11:09 09:38:58 (device local time).

Ultimately, the combined use of FTK Imager and ExifTool was necessary for the successful completion of the EXIF metadata analysis.

Q8: When was the file “Spending.xlsx” last opened by the user “phorger”? (If you used Autopsy to find the answer add a ‘*Bookmark*’ tag to the artifact that supports your answer-Provide the timestamps in UTC).

Answer:

The file 'Spending.xlsx' was last opened by the user 'phorger' at: 2024-12-24 22:57:58 UTC. It was located via the 'Recent Documents' section in the Autopsy results.

The screenshot shows the Autopsy 4.19.0 interface. The left sidebar displays the 'Data Sources' tree with 'BelkaCTF_6_CASE240405_LAPTOP.E01_1 Host' selected. The main window shows the 'Recent Documents' table with 32 results. The table has columns: Source Name, S, C, O, Path, Date Accessed, Data Source, and Comment. The row for 'Spending.xlsx' is highlighted, showing it was accessed on 2024-12-24 22:57:58 UTC. Below the table, the 'Spending.xlsx' artifact details are shown, including its path, date accessed, source file path, and artifact ID.

Source Name	S	C	O	Path	Date Accessed	Data Source	Comment
Junk				Y:\	0000-00-00 00:00:00	BelkaCTF_6_CASE240405_LAPTOP.E01	
Capture.Ink				C:\Users\phorger\Pictures\Capture.PNG	2024-01-05 17:25:37 UTC	BelkaCTF_6_CASE240405_LAPTOP.E01	
Music.Ink				C:\Users\phorger\Music	0000-00-00 00:00:00	BelkaCTF_6_CASE240405_LAPTOP.E01	
Videos.Ink				C:\Users\phorger\Videos	0000-00-00 00:00:00	BelkaCTF_6_CASE240405_LAPTOP.E01	
Desktop.Ink				C:\Users\phorger\Desktop	0000-00-00 00:00:00	BelkaCTF_6_CASE240405_LAPTOP.E01	
Pictures.Ink				C:\Users\phorger\Pictures	0000-00-00 00:00:00	BelkaCTF_6_CASE240405_LAPTOP.E01	
Downloads.Ink				C:\Users\phorger\Downloads	0000-00-00 00:00:00	BelkaCTF_6_CASE240405_LAPTOP.E01	
Documents.Ink				C:\Users\phorger\Documents	0000-00-00 00:00:00	BelkaCTF_6_CASE240405_LAPTOP.E01	
776AR-04U.PDF.Ink				Y:\776AR-04U.PDF	2024-01-15 13:00:59 UTC	BelkaCTF_6_CASE240405_LAPTOP.E01	
Spending.xlsx.Ink				Y:\Spending.xlsx	2024-12-24 22:57:58 UTC	BelkaCTF_6_CASE240405_LAPTOP.E01	

Type	Value	Source(s)
Path	Y:\Spending.xlsx	RecentActivity
Path ID	-1	RecentActivity
Date Accessed	2024-12-24 22:57:58 UTC	RecentActivity
Source File Path	/img_BelkaCTF_6_CASE240405_LAPTOP.E01/vol_vols/Users/phorger/AppData/Roaming/Microsoft/Windows/Recent/Spending.xlsx	
Artifact ID	-9223372036854775779	

Q9: What was the selected time zone of this system?

Answer:

According to the analysis of the SYSTEM file (TimeZoneInformation) via the Registry Explorer tool, the selected time zone for the system was Eastern Standard Time (UTC-5).

Q10: Which the 'Enabled Modules' of this case were when it was analyzed by me?

Answer:

The 'Ingest History' section with 'Enabled Modules' shows us which Autopsy tools were activated to search the disk image.

- **Recent Activity: This specific module was exclusively responsible for recovering the Microsoft Edge browsing history.**
- **Embedded File Extractor: Allowed Autopsy to 'open' compressed files in the background, giving us direct access to their contents.**
- **Hash Lookup: Was the necessary function that allowed the immediate identification of the image 'Capture2.png' via the known hashes database.**

Therefore, this directory proves which 'Data Artifacts' categories became available for examination, determining the scope of the forensic investigation.

Ingest History:

Job 1:

Data Source: BelkaCTF_6_CASE240405_LAPTOP.E01

Status: COMPLETED

Enabled Modules:

- Recent Activity
- Hash Lookup
- File Type Identification
- Extension Mismatch Detector
- Embedded File Extractor
- Picture Analyzer
- Keyword Search
- Email Parser
- Encryption Detection
- Interesting Files Identifier
- Central Repository
- PhotoRec Carver
- Virtual Machine Extractor
- Data Source Integrity
- Android Analyzer (aLEAPP)
- DJI Drone Analyzer
- YARA Analyzer
- iOS Analyzer (iLEAPP)
- GPX Parser
- Android Analyzer

Epilogue

This assignment highlighted in practice the complexity but also the methodicalness required during Digital Forensics. The combined use of Autopsy, FTK Imager, Registry Explorer and ExifTool, each targeting a different layer of the evidence, allowed the reconstruction of a complete forensic picture.